CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Memory Organization

Lecture No.- 1



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Recap of Previous Lecture



Topic DMA

Topic Modes of DMA

Topics to be Covered



- Topic: Memory Hierarchy
- Topic: Memory Presentation
- Topic: Memory Address Decoder
- Topic: Main Memory

[NAT]

GATE - PYQ

#Q. Consider a computer with a 4MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (in bits per second) of the DMA controller if 1% of the processor cycles are used for DMA?

A 2,56,000

B 3,200

C ✓ 25,60,000

D 32,000

$$t_y = 1 \text{ cycle} - \frac{1}{4\text{MHz}} = 0.25\,\mu\text{sec}$$

$$1\% = \frac{0.25\,\mu\text{sec}}{t_x} * 100\%$$

$$t_x = 25\,\mu\text{sec}$$

In 25 μsec, data = 8B = 64 bits

In 1 sec — " — $= \frac{64 \text{ bits}}{25 * 10^{-6}\text{sec}} = 2560000$ bits/sec

[NAT] GATE – PYQ

#Q. On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

```
        Initialize the address register      — 1
        Initialize the count to 500          — 1
LOOP:   Load a byte from device              — 2
        Store in memory at address given by address register — 2
        Increment the address register       — 1
        Decrement the count                  — 1
        If count !=0 go to LOOP              — 1
```

2 cycles

500 * 7 = 3500

3502 cycles

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speed up when the DMA controller-based design is used in a place of the interrupt driven program-based input-output?

$$\text{DMA time} = 20 + (500 * 2)$$
$$= 1020 \text{ cycles}$$

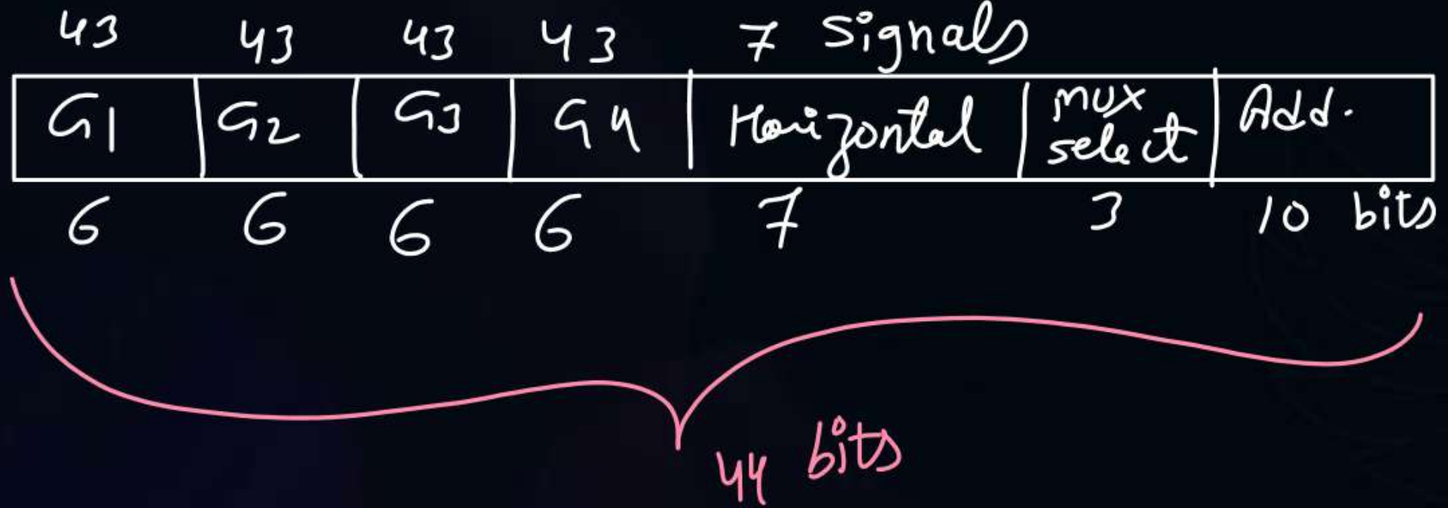
$$\text{speed up of DMA} = \frac{\text{interrupt I/O time}}{\text{DMA time}}$$

$$= \frac{3502}{1020}$$

$$= 3.4$$

[NAT]

#Q. Design of a vertical microprogrammed control unit requires to generate 50 signals. Out of first 43 those only 4 signals can be active at a time. And for remaining 7, anyone can be active anytime. The microinstruction of the control unit stores control signal information along with 3-bit mux select and 10-bits address field. The size of control memory required is?



control mem. size
$= 2^{10} * 44$ bits
$= 44$ K bits
Ans.



Topic : Memory Hierarchy

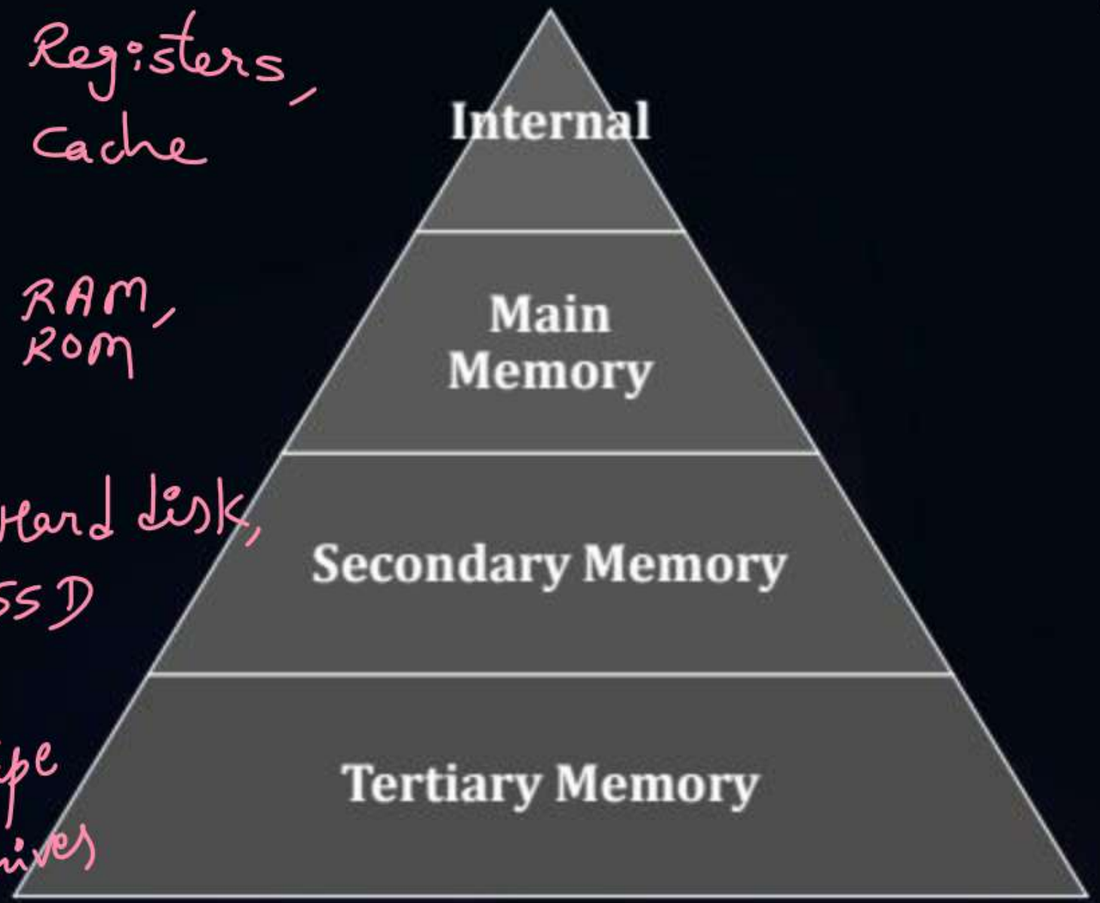
Memory hierarchy used when discussing performance issues.

Goal of Memory Hierarchy:

1. To maximize the Access Speed
2. To minimize the Per Bit Storage Cost



Topic : Memory Hierarchy



Access speed ↑

Per bit Storage cost ↑

Size ↓

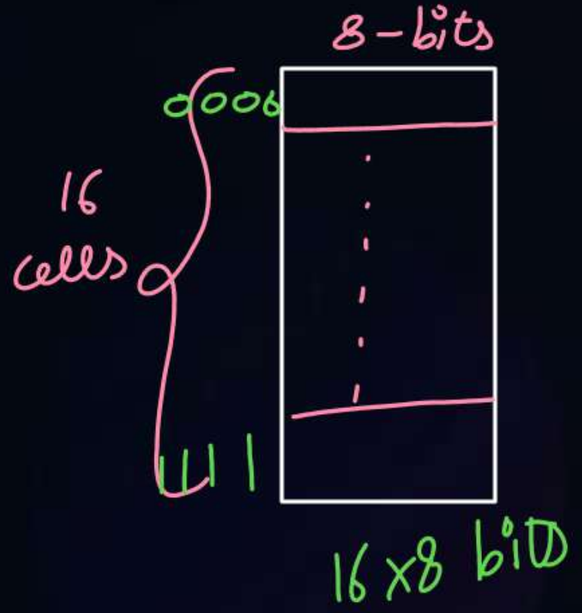
CPU Access frequency ↑



Topic : Memory Presentation

= No. of cells × 1 cell capacity

= No. of mem. addresses × bits per address





Topic : Memory Presentation

no. of cells	16	32	2^n	x
add.	4 bits	5 bits	n bits	$\lceil \log_2 x \rceil$ bits

Byte addressable mem.:-

ex:- 128K x 8 bits

or

128K x 1B

or

128KB

[MCQ]

#Q. Memory is represented as?

A. A×B where A = No. of memory locations, B = No. of bits in each location

B. $2^a \times B$ where a = No. of address bits, B = No. of bits in each location

C. B×A where, B = No. of bits in each location, A = No. of memory locations

D. ✓ (A) & (B) both

8×16

[MCQ]

#Q. A memory has 14-bits address bus. Then how many memory locations are there?

$2^{14} = 16K = 16384$

A. 16K

B. 16384

C. 2^{14}

D. ✓ All

Memory cycle:-

Time in which read or write is performed at an address in memory.

[MCQ]

#Q. The memory cycle time of a memory is 200nsec. The maximum rate with which the memory can be accessed?

Note: Consider memory as byte addressable.

A. 500 Bytes / Sec

B. 2000 Bytes / Sec

C. 5 Mbytes / Sec

D. 5 GBytes / Sec

in 200 ns, data = 1B

in 1 sec —11— $= \frac{1\ B}{200 * 10^{-9}\ sec}$

$= \frac{10^9\ B}{200\ sec}$

$= \frac{1000 * 10^6\ B}{200\ sec}$

$= 5\ MB/sec$

[NAT]

GATE PYQ

#Q. A processor can support a maximum memory of 4 GB, where the memory is word addressable (a word consists of two bytes). The size of the address bus of the processor is at least 31 bits?

$$\text{no. of cells} = \frac{4\text{GB}}{2\text{B}} = 2\text{G} = 2^1 \cdot 2^{30} = 2^{31}$$

$\Downarrow$

add. = 31 bits



2 mins Summary

Topic Memory Hierarchy

Topic Memory Presentation

Topic Memory Address Decoder

Topic Main Memory



Happy Learning

THANK - YOU